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MILITARY SPECIFICATION

SQUIB, ELECTRIC, M1A1, LOADING ASSEMBLING AND PACKING

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for one type of Squib, designated as the Squib, Electric, M1A1 for use in ammunition.

2. APPLICABLE DOCUMENTS

2.1 The documents of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN–P–223 — Powder, Black

MIL–P–116 — Preservations, Method of

MIL–G–2550 — General Specification for Ammunition Except Small Arms Ammunition

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection

MIL–STD–129 — Marking for Shipment and Storage

DRAWINGS

ORDNANCE CORPS

82–5–146 — Squib, Electric, M1A1, Assembling and Details

PUBLICATIONS

ORDNANCE CORPS

ORD–M608–11 — Sampling Tables and Procedures for Continuous Sampling by Attributes

(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be ob-

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tained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Squibs. The squibs shall comply with all requirements specified on the drawings, and with all requirements specified in applicable specifications.

3.3 Flash charge. The total volatile content of the flash charge shall not exceed, after loading, .80 percent when tested as specified in 4.4.1.

3.4 Resistance. The resistance of the bridge wire circuit of the loaded squib, including lead wires, shall be $1.00 \pm .25$ ohms when tested as specified in 4.4.4.

3.5 Pull test of loaded squib assembly. The loaded squib assemblies shall withstand a pull of 4 pounds minimum when tested as specified in 4.4.3.

3.6 Functioning. The loaded squibs shall function with an ignition delay not exceeding .10 seconds, (see 6.3) without misfire (see 6.4), and without any evidence of unburned black powder or any other malfunctioning, when tested as specified in 4.4.5.

3.7 Maximum safe current. No squib shall function when subjected to a direct current of 250 milliamperes for a minimum of 5 seconds, when tested as prescribed in paragraph 4.4.2.

3.8 Workmanship. All parts shall be free from chips, dirt, grease, corrosion, and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. A lot shall consist of squibs loaded in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Each lot shall be limited to:

(a) Flash charge composition from not more than one lot.

(b) Inert parts of one interfix lot number from one manufacturer.

(c) Material loaded in one day.

4.2 Sampling. Samples will be selected from each lot by the inspector as directed in the following subparagraphs.

4.2.1 Sample sizes and acceptance numbers for visual and dimensional attribute inspection will be selected in accordance with Standard MIL–STD–105 or ORD–M608–11, as appropriate. The contracting officer will indicate the set of tables appropriate to each inspection station, basing his choice upon the nature of the production line, availability of space and personnel, and other considerations.

4.2.2 *Functioning.* One hundred fifty squibs will be selected at random from each lot for the tests described in 4.4.5.

4.2.3 *Pull test of loaded squib assembly.* Fifty squib assemblies will be selected at random from each lot for the test described in 4.4.3.

4.2.4 All samples will be randomly selected proportionately from each production unit which is defined as the hourly production of each loading operator using the same batch of flash charge composition loaded under conditions of continuous production free from interruptions.

4.3 Inspection. Inspection will be performed by the inspector as specified in MIL–G–2550 and as indicated below. If a sample does not pass the inspection, the material from which the sample was taken will be rejected; how-

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ever, the contractor may screen out the defect or defects causing failure and resubmit the material if he so desires.

4.3.1 *Components and subassemblies.* Prior to assembly, it will be ascertained by reference to data cards, description sheets or other inspection documents that components and subassemblies of the squibs which are procured under separate specifications have been inspected, tested and accepted in accordance with their respective specifications and drawings.

4.3.2 *Classification of defects.*

4.3.2.1 *Assembly (prior to buttering the flash charge), (see Drawing (Dwg) 82–5–146A.*

Categories and defects	*Method of inspection*
Major AQL 0.25 percent	
1. Pull test of bridge wire joint	Approved testing
Minor	
None	

4.3.2.2 *Assembly (prior to varnish dip) (see Dwg 82–5–146A)*

Categories and defects	*Method of inspection*
Major AQL 0.25 percent	
1. Burr wire above outside surface of holder assembly	Visual
Minor AQL 1.50 percent	
41. Charge above flush or more than specified distance below end of sleeve	Gage
42. Sleeve not fully seated	Visual-Gage (see 4.3.3.1)
43. Cracks or perforations	Visual
44. Foreign matter	Visual

4.3.2.3 *Assembly (see Dwg 82–5–146A)*

Categories and defects	*Method of inspection*
Major AQL 0.25 percent	
1. Lead wires not twisted the required number of turns	Visual
Minor AQL 2.50 percent	
41. Wire insulation damaged	Visual
42. Length of insulated portion of lead wire, minimum (min.)	Scale
43. Length of bare portion of lead wire	Scale
44. Varnish inadequate	Visual (see 4.3.3.2)
45. Foreign matter	Visual

4.3.3 *Defect standards.*

4.3.3.1 Any assembly having a gap between the sleeve and the holder extending 360° around the circumference will be classed defective. However, if contact is made at any one point, then only those assemblies with a gap in excess of .015 inch will be classed defective.

4.3.3.2 Any assembly containing bubbles or an uncoated area on the sleeve or holder assembly, including the flash charge, will be classed defective.

4.4 Tests. The following tests shall be conducted by contractor personnel at the direction of and under the observation of the inspector.

4.4.1 *Test of flash charge.* The total volatile content of the flash charge shall be determined and the ingredients shall be verified in the following manner: A minimum of 3 samples each of 5 complete squib assemblies shall be selected at the start of each shift and an additional group of 3 samples shall be selected after the lunch period. Two of the three samples of each group shall be held in a suitable safe storage place to be used when a sample is rejected (see 4.4.1.3). The third sample shall be analyzed as described in the following paragraphs.

4.4.1.1 *Total volatiles.* An accurately weighed portion of approximately 1 gm. of the flash charge composition shall be transferred to a tared moisture dish two inches in diameter. The moisture dish and contents shall be placed in a drying oven at 55°—

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65°C. for two hours, cooled in a desiccator and weighed. The loss in weight of the dish and contents shall be calculated to percent total volatiles in the sample.

$$\text{Percentage of total volatiles} = \frac{100A}{W}$$

where:

A = loss in weight of moisture dish and contents in grams

W = weight of sample in grams.

4.4.1.2 *Composition.*

4.4.1.2.1 *Diazodinitrophenol.* An accurately weighed portion of approximately one gm. of the sample shall be transferred to a 250-ml. beaker and 150-ml. of ethylene chloride added. The beaker and contents shall be heated on a steam bath for two hours and stirred occasionally. The contents of the beaker shall be filtered with the aid of a tared medium-sized-porous-glass crucible and suction. The beaker and residue in the crucible shall be washed with boiling ethylene chloride until the filtrate coming through the crucible is no longer colored yellow. The crucible and residue shall be dried in an oven maintained at 80° ± 5°C. for two hours, cooled in a desiccator and weighed. The percentage of diazodinitrophenol shall be calculated in the sample on a volatile-free basis as follows:

$$\text{Percentage of Diazodinitrophenol} = \frac{(A-B-C)\ 100}{W\ D}$$

where:

A = weight of glass crucible and sample in grams

B = weight of glass crucible and residue in grams

C = weight of volatile matter in sample

W = weight of the sample in grams on a volatile-free basis

D = percent solubility of diazodinitrophenol in ethylene chloride expressed as a decimal fraction obtained as follows:

A 0.1900 ± .0010 gm. of dry diazodinitrophenol from the lot used in the manufacture of the flash charge composition being analyzed shall be placed in a 250-ml. beaker and 150-ml. of ethylene chloride added. The beaker and contents shall be heated on a steam bath for two hours and stirred occasionally. The contents of the beaker shall be filtered with the aid of a tared medium-sized-porous-glass crucible. The beaker and residue in the crucible shall be washed with hot ethylene chloride until the filtrate coming through the crucible is no longer yellow in color. The crucible and residue shall be dried in an oven maintained at 100° ± 5°C. for one hour, cooled in a desiccator and weighed. The loss in weight of the crucible and contents shall be calculated to percent solubility of diazodinitrophenol in ethylene chloride. This value is represented by the letter "D" given in the equation for calculating the percentage of diazodinitrophenol. A value of 0.9726 shall be used except in case of non-compliance, dispute or question. In such cases, the value determined as described above shall be used.

4.4.1.2.2 *Nitrostarch.* The dry residue obtained in the above determination shall be extracted with ten five-ml. portions of hot water, stirring the residue in the crucible after the addition of each portion before applying suction. The crucible and residue shall be dried in an oven maintained at 80° ± 5°C. for two hours, cooled in a desiccator and weighed. The dry residue obtained after the water extraction of the potassium chlorate shall be extracted with ten five-ml. portions of hot acetone, the residue in the crucible shall be stirred after the addition of each portion before applying suction. The crucible and residue shall be dried in an oven maintained at 100° ± 5°C, for two hours, cooled in a desiccator and weighed. The percentage of nitrostarch in the sample shall be calculated on a volatile-free basis as follows:

$$\text{Percentage of Nitrostarch} = \frac{100\ (A-B)}{W}$$

where:

A = weight of glass crucible and residue

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before extraction with acetone in grams

B = weight of glass crucible and residue after extraction with acetone in grams

W = weight of sample in grams on a volatile-free basis

4.4.1.2.3 *Powdered wood charcoal.* From the weight of the glass crucible and residue after extraction with acetone and the weight of the glass crucible the percentage of powdered wood charcoal shall be calculated on a volatile-free basis as follows:

$$\text{Percentage of wood charcoal} = \frac{100\ (A-B)}{W}$$

where:

A = weight of glass crucible and residue after extraction with acetone in grams

B = weight of glass crucible in grams

W = weight of the sample in grams on a volatile-free basis

4.4.1.2.4 *Potassium chlorate.* The percentage of potassium chlorate in the sample shall be calculated by subtracting from 100 the combined percentages of diazodinitrophenol, nitrostarch, and powdered wood charcoal.

4.4.1.3 Rejection and retest. If a sample fails to pass the above tests, that portion of the shift's production from which the sample was drawn will be rejected. A retest will be permitted only if the validity of the test method is in question. When a retest is indicated, the procedure is as follows: The two remaining samples, (see 4.4.1) of the group in question, shall be subjected one sample at a time to the tests listed in paragraphs 4.4.1.1 thru 4.4.1.2.4. If one more sample fails to pass the test, the validity of the test method is satisfied and that portion of the shift's production represented by the failing samples will be rejected. If on the other hand the two remaining samples pass the tests, the questioning of the validity of the test method is justified and that portion of the shift's production from which the samples were drawn will be considered usable.

4.4.2 *Test for maximum safe current.* A sample of one hundred fifty squib assemblies from each lot shall be tested as follows: Each squib in the sample shall be subjected to a direct current of 250 + 10 milliamperes for a minimum of five seconds.

4.4.2.1 *Rejection and retest.* If two or more squibs in the sample function, the lot shall be rejected not subject to a retest. If not more than one squib in the sample functions, the lot may be retested using double the sample size. The lot will be rejected if two or more squib function in the combined test and retest.

4.4.3 *Pull test of loaded squib assembly.* The sample squib assemblies shall be assembled in a satisfactory fixture. A load of four pounds minimum shall be applied along the longitudinal axis of each lead wire for a period of 15 ± 2 seconds. The squib assemblies shall then be tested for resistance in accordance with 4.4.4.

4.4.3.1 *Retest and rejection.* If not more than one squib assembly fails to meet the requirements of 3.4 or 3.5, the lot may be retested at the request of the contractor. The sample size for retest will be 100 squib assemblies. If more than one squib assembly fails in the combined test and retest, the lot will be rejected.

4.4.4 *Resistance test.* Each squib in the lot shall be tested for resistance by means of a suitable resistance-measuring instrument value of 0.9726 shall be used except in case having a short circuit current of 10 milliamperes, maximum. Observe each squib to determine compliance with 3.4.

4.4.4.1 *Rejection.* Any squib that does not meet the requirement of 3.4 shall be rejected.

4.4.5 *Functioning.* The sample unconditioned loaded squib assemblies shall be tested as follows:

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4.4.5.1 *Procedure.* The squib assemblies shall be embedded in ten grams of Grade A–5 black powder, Specification JAN–P–223 contained in a small cloth bag. Prior to connection of the squib assemblies to the test circuit, the circuit shall be prepared for firing by adjusting the current through the one ohm precision resistor (see figure 1) to 545 ± milliamperes. Following preparation of the test circuit, the assemblies shall be fired behind a suitable barricade in accordance with figure 1.

4.4.5.2 *Retest and rejection.* If one or two squib assemblies fail to meet the requirements of 3.6, the lot may be retested at the request of the contractor. The sample size for retest will be 300 squib assemblies. If more than two squib assembly fail in the combined test and retest the lot will be rejected.

4.4.6 *Acceptance.* If the requirements of the above inspection and tests have been met, and if the inspector is assured that all other requirements have been met, the lot will be accepted.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level A.* Two hundred (200) squibs, each with the wire extended straight shall be packed in a set-up paperboard box that complies with the requirements of Specification JAN–P–133. All openings of the set-up box shall be sealed with gummed paper tape conforming to the requirements of Specification UU–T–111. The packed squibs shall be sealed in a Type II Class e bag complying with the requirements of Specification MIL–B–117.

5.2 Packing.

5.2.1 *Level A.* Squibs packaged as specified in paragraph 5.1.1 shall be packed in quantities not to exceed 50 pounds net weight in a sealed waterproof case liner within a type 2 or 4 nailed wood box constructed and strapped in accordance with the requirements of Specification JAN–P–106. The waterproof liner shall conform with the requirements of Specification MIL–B–13239 for Grade B Class II type CW–1 or CW–2 material.

5.2.2 *Level C.* Squibs packaged as specified in paragraph 5.1.1 shall be packed in quantities not to exceed 30 pounds net weight in a FSC 275 pound minimum test corrugated fiberboard carton. The carton shall comply with the requirements of Specification LLL–B–631. The carton shall be sealed with gummed paper tape not less than two inches wide conforming to the requirements of Specification UU–T–111 for Class 2 material.

5.3 **Marking.** Packages and packs of squibs shall be marked in accordance with the requirements of Specification MIL–STD–129A and applicable I.C.C. regulations.

5.4 **Data cards.** Data card information shall be as specified in Specification MIL–G–2550.

6. NOTES

6.1 Specification JAN–P–223 listed in 2.1 is used only in connection with test of this item.

6.2 Specification MIL–P–116 listed in 2.1 is used only in connection with packaging of this item.

6.3 **Ignition delay.** An ignition delay is time elapsed from initiation of the current to rupture of the squib bridge wire.

6.4 **Misfire.** A misfire is a failure of the squib to function due to defects within the squib.

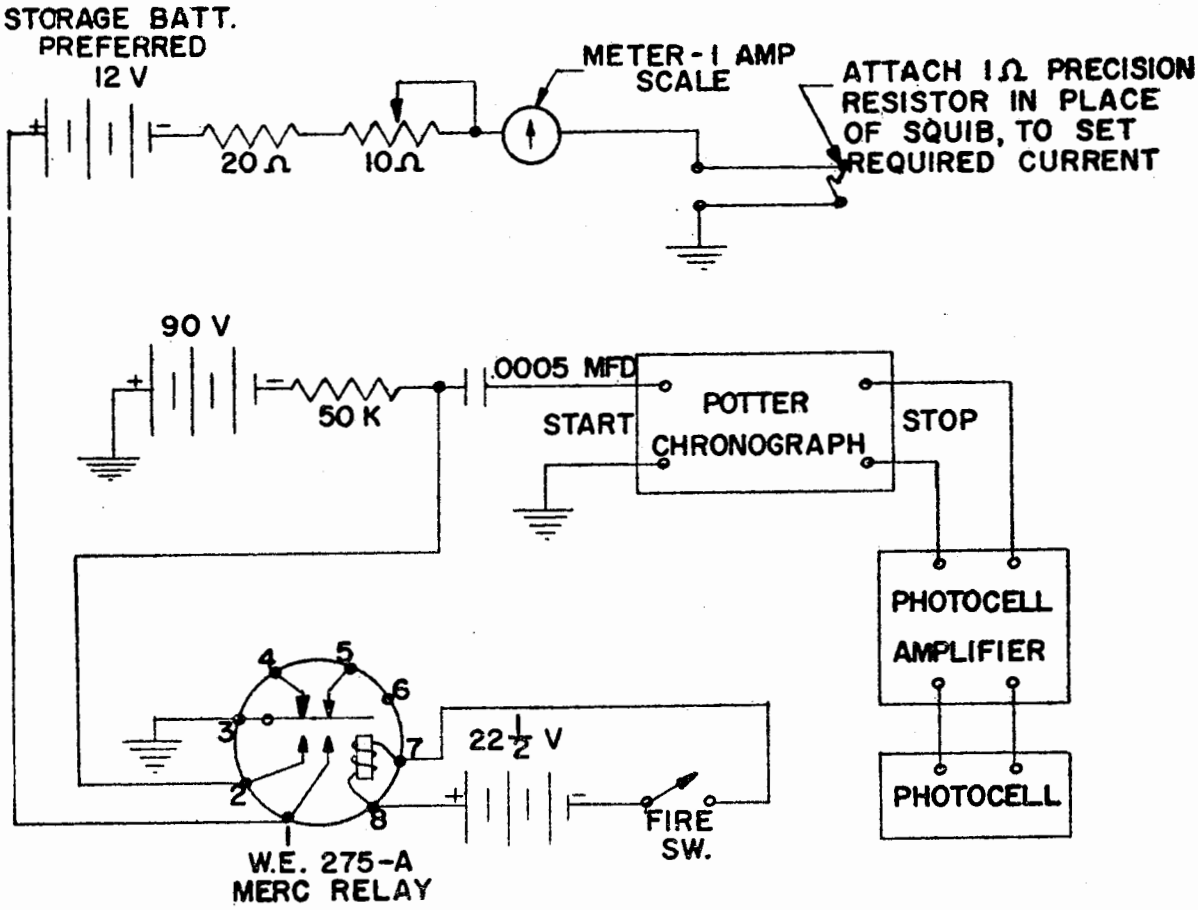
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NOTE:-

THE CIRCUIT MUST BE AS INDICATED SO THAT VARIATION OF SQUIB RESISTANCE WITHIN LIMITS OF SPECIFICATIONS WILL NOT VARY FIRING CURRENT BY MORE THAN APPROX 1%.

W.E. 275-A MERC. RELAY MUST BE USED BECAUSE OF SIMULTANEITY OF CONTACT CLOSURE.

PHOTO TUBE SHOULD HAVE S-1 SPECTRAL RESPONSE.

FIGURE 1

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